Request: new dataset + service account — Market Research Agent

Project: `ik-marketing-data` **Requested by:** india.sales@interviewkickstart.com **Workload:** Market-research agent pipeline (AI/ML India). First table live: monthly LinkedIn **job-market demand**. The same dataset will later hold the other tabs (social trends, blog/publications, competitor intel, executive summary) + back a frontend dashboard.

This follows the per-workload least-privilege pattern in `Call Analysis Intelligence (CAI)/docs/SERVICE_ACCOUNT_AUDIT.md`.

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1. Create one dataset

	Value
Dataset ID	`Market_Research`
Project	`ik-marketing-data`
Location	**US** (match existing datasets)

> Pre-create it so the SA never needs project-level `datasets.create`. The pipeline uses `create_table(exists_ok=True)`, so it won't try to make the dataset itself.

2. Create one service account

	Value
Name	`market-research-bq`
Email	`market-research-bq@ik-marketing-data.iam.gserviceaccount.com`
Used by	the monthly scraper on Christo's machine (and future tabs of the same project)

3. Grant exactly these roles (least privilege)

Scope	**Role**	**Why**
Project `ik-marketing-data`	`roles/bigquery.jobUser`	run load / query / delete jobs
Dataset `Market_Research`	`roles/bigquery.dataEditor`	create tables, append snapshots, idempotent `DELETE WHERE run_id=…`, read-back

OAuth scopes (set in code): `bigquery`, `cloud-platform`. **No Drive scope** (no Sheets-federated tables).

4. What it does NOT need (please don't grant)

- ❌ `roles/bigquery.admin` — too broad

- ❌ any read/write on `India_Leads`, `Marketing_data_new_logic`, or other upstream datasets — this pipeline reads **only** from Apify, never from BQ
- ❌ `datasets.create` / `datasets.delete` / `setIamPolicy` — dataset is pre-created; never mutates IAM
- ❌ `routines.create`, `connections.create`

This SA is strictly write-scoped to one dataset + job execution. Total compromise leaks only public job-posting aggregates (no PII).

5. Key delivery

Issue a JSON key and hand it to Christo. **This workload runs on Fly.io**, so the key is consumed as a **Fly secret** (base64-encoded), not a file on a server:

```
fly secrets set MARKET_RESEARCH_BQ_SA_B64="$(base64 -w0 market-research-bq.json)"
\
  -a ik-market-research-jobs
```

(A copy may also sit at `C:\Users\Admin\.gcp-keys\market-research-bq.json` for local runs, same convention as the other 6 SAs.) 90-day rotation, standard key handling — never commit it.

6. One-shot commands (if the team prefers CLI)

```
# dataset
bq --location=US mk --dataset ik-marketing-data:Market_Research

# service account
gcloud iam service-accounts create market-research-bq \
  --project=ik-marketing-data \
  --display-name="Market Research Agent (BQ writer)"

# project: jobUser
gcloud projects add-iam-policy-binding ik-marketing-data \
  --member="serviceAccount:market-research-bq@ik-marketing-
data.iam.gserviceaccount.com" \
  --role="roles/bigquery.jobUser"

# dataset: dataEditor — easiest in console:
#   BigQuery → Market_Research → Sharing → Permissions →
#   Add principal: market-research-bq@ik-marketing-data.iam.gserviceaccount.com
#   Role: BigQuery Data Editor

# key
gcloud iam service-accounts keys create market-research-bq.json \
  --iam-account=market-research-bq@ik-marketing-data.iam.gserviceaccount.com
```

7. Smoke test (proves the grant is correct)

```
from google.cloud import bigquery
from google.oauth2 import service_account
creds = service_account.Credentials.from_service_account_file(
    r"C:\Users\Admin\.gcp-keys\market-research-bq.json",
    scopes=["https://www.googleapis.com/auth/bigquery",
            "https://www.googleapis.com/auth/cloud-platform"])
c = bigquery.Client(credentials=creds, project="ik-marketing-data")
c.query("CREATE TABLE IF NOT EXISTS `ik-marketing-
data.Market_Research._smoke_DELETE_ME`(x INT64)").result()
c.query("INSERT INTO `ik-marketing-data.Market_Research._smoke_DELETE_ME`(x)
VALUES(1)").result()
c.query("DROP TABLE `ik-marketing-
data.Market_Research._smoke_DELETE_ME`").result()
print("OK — dataEditor + jobUser confirmed")
```

Expect all three to succeed. A `PERMISSION_DENIED` means a grant is missing — escalate before closing.

After the key is delivered

Going live on the real dataset is two env vars (no code change):

```
$env:JOBMKT_BQ_KEY     = "C:\Users\Admin\.gcp-keys\market-research-bq.json"
$env:JOBMKT_BQ_DATASET = "Market_Research"
```

Optional — separate read-only SA for the dashboard

When the frontend dashboard is built, give it its **own** read-only identity rather than reusing the writer key, so the dashboard can never mutate the table:

	Value
Name	`market-research-dash-bq`
Email	`market-research-dash-bq@ik-marketing-data.iam.gserviceaccount.com`
Project `ik-marketing-data`	`roles/bigquery.jobUser`
Dataset `Market_Research`	`roles/bigquery.dataViewer` (read-only)